OPOKU CLAUDIOUS SAMUEL MENSAH

Geological Engineer
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SUMMARY

Highly motivated and skilled Geological Engineering graduate with hands-on experience in water resources, infrastructure research, and mobile development. Proficient in remote sensing and GIS applications, particularly in environmental monitoring and impact assessment. Currently enhancing software development skills through national service, with a proven ability to integrate geological knowledge and technology.

EXPERIENCE

Mobile App Developer

Mar '25 — Mar'26

Ayzen Engineering LTD, Accra

- Developing and maintaining high-performance mobile applications using Flutter, delivering seamless experiences across both Android and iOS platforms. Designed and implemented intuitive, responsive user interfaces aligned with modern UI/UX standards to improve usability and accessibility. Integrated RESTful APIs and backend services to enable real-time data flow, secure authentication, and smooth in-app interactions. Utilized state management solutions such as Riverpod, Bloc, Provider, and GetX to handle complex logic, data synchronization, and scalable app architecture. Played a key role in developing and enhancing the Ghana Institute of Geoscientists (GhIG) mobile application, ensuring smooth navigation, optimized performance, and alignment with organizational needs. Performed debugging, profiling, and performance optimization to reduce load times, improve responsiveness, and ensure app stability in production environments. Collaborated closely with engineers, designers, and project leads to deliver reliable digital solutions for the company's engineering and professional service platforms.

Web Developer

Sept '24 — Mar'25

IKOLILU, Accra

- Developed and maintained scalable web applications and dashboards to support the company's educational technology platform. Built responsive and user-friendly interfaces using Next.js, React, TypeScript, and Tailwind CSS, ensuring consistency across devices and browsers. Integrated backend services and third-party APIs to deliver real-time data, authentication, and seamless user interactions within the platform. Implemented efficient state management solutions using Context API, Redux, and React Query to optimize data fetching and application performance. Collaborated with product and design teams to translate requirements into functional features, improving overall user experience and platform usability.
  Conducted thorough testing, debugging, and performance optimization to enhance loading speed, accessibility, and system reliability. Followed best practices in code quality, version control (Git/GitHub), and deployment workflows to ensure maintainability and smooth updates.

National Service Personnel – Mobile App Developer

Nov '23 — Sept'24

Really Great Tech, Accra

- Developing and maintaining cross-platform mobile applications using Flutter for both iOS and Android platforms.

Designed and implemented responsive user interfaces that provide excellent user experience across different devices and screen sizes. Integrated RESTful APIs to fetch and display data within the app, ensuring seamless data synchronization and real-time updates. Utilized various state management techniques such as Provider, Riverpod, Bloc, and GetX to manage application state effectively.
Implemented state management solutions to handle complex UI interactions and data flows within the app. Conducted performance testing and optimization to improve app responsiveness and reduce load times. Implemented efficient coding practices, including lazy loading, caching, and memory management, to enhance app performance.

Ghana Water Company - Intern Nov '21 — Dec '21
Ghana Water Company, Mampong Ashanti Mampong Ashanti, Ghana

- Assisted in the collection and analysis of water samples to ensure compliance with quality standards.
  Conducted various chemical and microbiological tests to assess water quality, including pH, turbidity, and bacterial contamination. Monitored the performance of water treatment processes, including filtration, sedimentation, disinfection, and distribution. Collected and analyzed data on water flow rates, chemical dosages, and treatment efficiency. Conducted site visits to water treatment plants and distribution facilities to observe operations and identify potential issues.

Building and Road Research Institute – Intern Nov'22 — Dec '23
Fumesua, Ghana

- Assisted in collecting geotechnical and structural data from various construction sites and road projects.
  Conducted soil sampling, testing, and analysis to determine soil properties and suitability for construction. Managed and organized data collected from field and laboratory tests using software tools such as Excel. Prepared charts, graphs, and presentations to summarize research findings and project progress. Assisted in the preparation of presentations for stakeholders, including government agencies and industry partners.

Programming Languages JavaScript, Dart, Typescript
Database Management Firebase
Remote Sensing Tool Google Earth Engine
GIS Tool QGIS

PROJECTS

Mining Impact Monitoring in Obuasi

Final Year Project, Kwame Nkrumah University of Science and Technology, Feb, 24 – Aug, 24
During my final year at KNUST, I undertook a comprehensive project to monitor and assess the environmental impact of mining activities in Obuasi. The project aimed to utilize advanced remote sensing and GIS technologies to analyze land use changes, vegetation loss, and water quality deterioration due to mining. I used Google Earth Engine for processing and analyzing large datasets of satellite imagery, QGIS for spatial data analysis, visualization, and mapping.

Blockchain Hackathon Project – No-Loss Raffle (Base Batches 002 Program) 2025

Built a decentralized no-loss raffle application as part of the Base Batches 002 global hackathon, enabling users to participate in raffles without losing their principal deposit. Integrated Aave's lending protocol to deposit participant funds and generate yield, using the accrued interest as the prize pool while ensuring users maintain their original stake. Developed secure and gas-efficient smart contracts using Solidity and Foundry, with mechanisms for trustless prize distribution and transparent interaction flows. The project was selected as one of the Top 50 projects globally, highlighting its innovation, technical depth, and real-world applicability.

NDVI Change Detection Web App — Google Earth Engine

Developed an interactive geospatial analytics application using Google Earth Engine to analyze vegetation changes over time. The tool integrates Sentinel-2 imagery, Cloud Score+ cloud masking, NDVI computation, and median composites to visualize and compare vegetation health between two selected years. Implemented a side-panel UI with dynamic dropdown filters, a dual-map split-view for side-by-side comparison, and automated spatial filtering using GAUL administrative boundaries.

Project Link: https://ee-claudiouso00.projects.earthengine.app/view/ndvi-ashanti-region

LEADERSHIP

Geological Engineering President, 2021 - 2022
Kwame Nkrumah University of Science and Technology, Obuasi Campus

CERTIFICATION

Blockchain Basics Course, Cyfrin Updraft

EDUCATION

Geological Engineering, Kwame Nkrumah University of Science and Technology — Sep '19 — Aug '23
Kumasi, Ghana